PYQ Hub: A Centralized Previous Year Question Paper Portal

Branch-wise, year-wise and subject-wise exam paper repository

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Introduction

- Previous year question papers (PYQs) are the most trusted exam-prep resource
- Today they are scattered across WhatsApp groups, drives and seniors' phones
- We propose a web platform that organizes them in one searchable place

Problem Statement

- No single, reliable source for past papers
- Papers are unorganized, duplicated, low quality or missing
- Juniors depend on personal contacts, wasting time before exams
- No verification of paper authenticity or completeness

Objectives

- Build a centralized, easy-to-navigate PYQ repository
- Enable browsing via Branch $\rightarrow$ Year $\rightarrow$ Subject $\rightarrow$ Papers
- Allow students to upload and download papers with moderation
- Provide fast search and a clean, mobile-friendly experience

Literature Survey

Existing Approach	Limitation
WhatsApp / Telegram groups	Unorganized; older files get buried and lost
Google Drive folders	Inconsistent naming, no search or moderation
Commercial sites (Studocu, Course Hero)	Paywalled and not specific to our college
University websites	Often outdated with incomplete archives

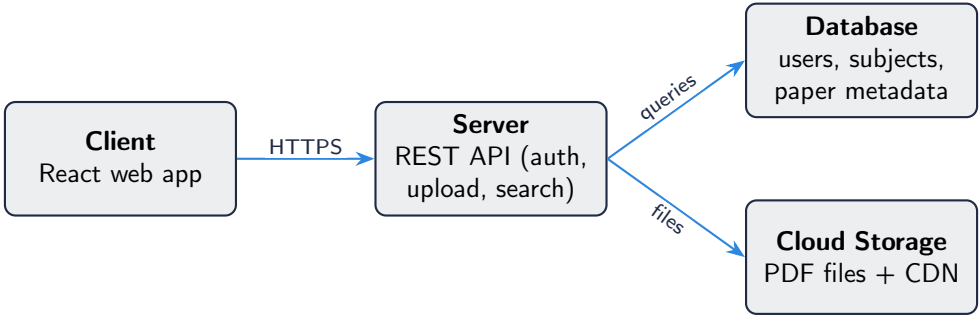
Research Gap

- No college-specific, structured and free PYQ platform
- Existing sources lack a branch / year / subject hierarchy
- No quality control, versioning or duplicate handling
- Weak search and no community contribution workflow

Proposed System

- Web app with navigation: Branch $\rightarrow$ Year $\rightarrow$ Subject $\rightarrow$ Content
- Student login, paper upload with admin approval, preview and download
- Search and filters by subject code, year and exam type (mid-sem / end-sem)
- Admin dashboard for moderation and usage analytics

System Architecture



Students access the portal through the browser; the server handles all logic and data access.

Methodology

- Requirement gathering through student surveys
- Agile, iterative development in short sprints
- Design $\rightarrow$ Implementation $\rightarrow$ Testing $\rightarrow$ Deployment
- Pilot with one batch and refine using their feedback

Modules

Module	Function
User Authentication	Sign-up, login and college email verification
Browse Module	Branch $\rightarrow$ Year $\rightarrow$ Subject navigation
Upload & Moderation	Students upload PDFs; admins approve or reject
Search & Filter	Find papers by subject code, year and exam type
Admin Dashboard	Manage content, users and view usage statistics

Technologies

Layer	Technology
Frontend	HTML , CSS
Backend	Node.js with Express (or Django / Flask)
Database	MongoDB / PostgreSQL
File Storage	Firebase Storage / AWS S3
Hosting	Vercel / Render
Authentication	JWT with college email verification

Expected Results

- A single trusted source of organized PYQs for all branches
- Less time spent hunting for study material
- A growing, community-driven and verified paper database
- Better exam preparation and student collaboration

Applications

- Semester exam preparation for students
- Reference for faculty to analyze paper trends
- Extendable to other colleges and universities
- Future scope: notes sharing and AI-based topic analysis

Timeline

Weeks	Activity	Deliverable
1	Requirement analysis, UI design	Wireframes, feature list
2	Backend, database, authentication	Working API
3	Frontend, upload and search modules	Functional web app
4	Testing and bug fixing	Test report
5	Deployment and pilot launch	Live portal

References

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[2] Express.js Documentation. https://expressjs.com

[3] MongoDB Documentation. https://www.mongodb.com/docs